

Time and Work — All Models with Formulas & Examples

Core Concept

If a person/machine takes **n days** to finish a work, then: **Work done in 1 day = $1/n$** (this is called the "rate of work")

Total work is usually assumed as **1 unit**, or taken as **LCM of given days** (makes calculations easier with whole numbers).

Model 1: Basic Work Formula

Formula:

Work = Time $\times$ Rate of work
Rate of work = $1/\text{Time taken to complete work}$

Example: A can finish a work in 10 days. What is A's 1-day work? $\rightarrow$ A's 1-day work = $1/10$

If A works for 4 days, work done = $4 \times 1/10 = 4/10 = 2/5$

Model 2: Two or More People Working Together

Formula:

If A takes 'a' days and B takes 'b' days, working together they take: **Time (together) = $(a \times b)/(a + b)$ days**

Combined 1-day work = $1/a + 1/b$

Example: A can do a work in 12 days, B can do it in 15 days. In how many days will they finish it together?

1-day work of A = $1/12$, of B = $1/15$
Combined 1-day work = $1/12 + 1/15 = 5/60 + 4/60 = 9/60 = 3/20$

Time together = $20/3 = 6\frac{2}{3}$ days

(Shortcut using formula: $(12 \times 15)/(12 + 15) = 180/27 = 20/3$ days $\checkmark$)

Model 3: Three People Working Together

Formula:

Combined 1-day work = $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ Time = $1 \div (\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$

Example: A, B, and C can complete a work in 10, 15, and 30 days respectively. Find the time taken together.

Combined 1-day work = $\frac{1}{10} + \frac{1}{15} + \frac{1}{30} = \frac{3}{30} + \frac{2}{30} + \frac{1}{30} = \frac{6}{30} = \frac{1}{5}$

Time = **5 days**

Model 4: Using the LCM Method (Efficiency Method) — Fastest for Exams

Steps:

Take total work = LCM of all given days

Efficiency (work per day) = Total work $\div$ Days taken

Add efficiencies for combined work

Time = Total work $\div$ Combined efficiency

Example: A takes 8 days, B takes 12 days. Together, how many days?

Total work = LCM(8, 12) = 24 units A's efficiency = $24/8 = 3$ units/day B's efficiency = $24/12 = 2$ units/day

Combined efficiency = 5 units/day

Time = $24/5 = 4.8$ days

Model 5: Work and Wages (Sharing of Money Based on Work Done)

Formula:

Wages are distributed in the ratio of **work done**, which equals the ratio of their **efficiencies** (or ratio of 1-day work), when working the same number of days. Share of A : Share of B = Efficiency of A : Efficiency of B

Example: A and B can do a work in 6 days and 8 days respectively. They work together and earn ₹840. Find each one's share.

Efficiency ratio A : B = $1/6 : 1/8 = 4 : 3$ (multiplying by 24)

A's share = $840 \times \frac{4}{7} = ₹480$ B's share = $840 \times \frac{3}{7} = ₹360$

Model 6: A Alone Takes 'x' Days More/Less Than Together

Formula (when B is x times as efficient as A, or takes fraction of time):

If A takes 'a' days alone, and B is k times as efficient as A, then B takes a/k days.

Example: A is thrice as efficient as B, and together they take 15 days. Find the time A alone will take.

Efficiency ratio A : B = 3 : 1 Combined efficiency = 4 units Total work = $4 \times 15 = 60$ units

A's efficiency = 3 units/day A's time alone = $60/3 = 20$ days

Model 7: Work Done in Alternate Days / Part-Time Work

Formula:

Calculate 1-day work of each, then add day-by-day work based on the pattern given (alternating, or one after another) until total work = 1 (or LCM value).

Example: A can finish work in 6 days, B in 8 days. They work on alternate days starting with A. In how many days will the work finish?

Total work = LCM(6,8) = 24 units A's efficiency = 4 units/day, B's efficiency = 3 units/day

Day 1 (A): 4 units → Total = 4 Day 2 (B): 3 units → Total = 7 Day 3 (A): 4 units → Total = 11 Day 4 (B): 3 units → Total = 14 Day 5 (A): 4 units → Total = 18 Day 6 (B): 3 units → Total = 21 Day 7 (A): 4 units → Total = 25 (exceeds 24, so work finishes partway through Day 7)

After Day 6: 21 units done, 3 units remaining Day 7 (A) needs: $3/4$ day to finish remaining 3 units

Total time = $6 + 3/4 = 6\frac{3}{4}$ days

Model 8: Some People Leave/Join Midway

Formula:

Break the problem into phases. Calculate work done in each phase separately, then solve for the remaining work/time in the next phase.

Example: A and B together can complete a work in 12 days. They work together for 4 days, then A leaves. B finishes the remaining work alone in 24 days. Find B's alone time to do the whole work.

Combined 1-day work (A+B) = $1/12$ Work done in 4 days = $4/12 = 1/3$

Remaining work = $1 - 1/3 = 2/3$ This $2/3$ work is done by B alone in 24 days So B's 1-day work = $(2/3)/24 = 2/72 = 1/36$

B alone takes **36 days** for the whole work.

Model 9: Pipes and Cisterns (Time & Work with Inlet/Outlet)

Formula:

Inlet pipe fills tank $\rightarrow$ positive work ($1/\text{time}$) Outlet pipe empties tank $\rightarrow$ negative work ($-1/\text{time}$) Net work = Sum of all pipes' rates (with correct sign)

Example: Pipe A can fill a tank in 6 hours. Pipe B can empty it in 9 hours. If both are opened together, how long to fill the tank?

A's rate = $+1/6$ (filling) B's rate = $-1/9$ (emptying)

Net rate = $1/6 - 1/9 = 3/18 - 2/18 = 1/18$

Time to fill tank = **18 hours**

Model 10: Men-Days-Hours (Work Equivalence Formula)

Formula (Most important for TET/competitive exams):

$M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2$ Where M = number of men, D = number of days, H = hours per day, W = amount of work

Example: 12 men working 8 hours a day can complete a work in 10 days. How many men are needed to complete the same work in 6 days, working 10 hours a day?

$M_1 = 12, D_1 = 10, H_1 = 8, W_1 = 1$ $M_2 = ?, D_2 = 6, H_2 = 10, W_2 = 1$

$12 \times 10 \times 8 / 1 = M_2 \times 6 \times 10 / 1$ $960 = 60 \times M_2$ $M_2 = 960/60 = \mathbf{16 \text{ men}}$

Model 11: Work Involving Men, Women, and Children (Different Efficiencies)

Formula:

Establish equivalence: e.g., if 2 men = 3 women in efficiency, convert everyone into a single unit (like "man-days") before solving.

Example: If 2 men or 3 women can do a piece of work in 10 days, how many days will 4 men and 6 women take to finish it together?

2 men = 3 women (equal efficiency) $\rightarrow$ 1 man = 1.5 women So 4 men = 6 women (in efficiency)

Total work = 2 men $\times$ 10 days = 20 man-days (using men as unit) Since 4 men = 6 women, "4 men + 6 women" = "4 men + 4 men" (converting women to men) = 8 men

Time = 20 man-days / 8 men = **2.5 days**

Model 12: Work Efficiency Percentage Comparison

Formula:

If A is x% more efficient than B, then: Ratio of times taken (A:B) = 100 : (100+x) Ratio of efficiencies (A:B) = (100+x) : 100

Example: A is 25% more efficient than B. If B takes 25 days to complete a work, find A's time.

Efficiency ratio A:B = 125:100 = 5:4 Time ratio A:B = 4:5 (inverse of efficiency)

B's time = 25 days corresponds to ratio 5 A's time corresponds to ratio 4 A's time = $25 \times \frac{4}{5} = \mathbf{20 \text{ days}}$

Quick Reference Table

Scenario	Formula
1-day work	$1/\text{Time}$
Two people together	$(a \times b)/(a+b)$
Wage sharing	Ratio of efficiencies
Pipes (inlet+outlet)	Sum of rates (inlet +, outlet -)
Men-Days-Hours	$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$
% more efficient	Time ratio = 100 : (100+x)

Tip for TET prep: Most Time & Work problems become simple once you convert everything into "efficiency" (work per day) using the **LCM method** — it avoids fractions and reduces calculation errors under time pressure.